The Social and Cultural Effects of Markets:
Evidence from Folklore and Ethnography

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The social and cultural consequences of markets have long been debated. Part of the difficulty in empirically investigating this topic lies in obtaining data that cover an array of societies with diverse social, cultural, and economic configurations. Folklore and ethnography provide this data. Folklore reflects a people's internal worlds and external environments from an insider perspective. Ethnography depicts a people from an outsider perspective. I employ folklore text from the Berezkin catalogue of folklore motifs and ethnographic text from the HRAF World Cultures database to analyze the social and cultural effects of markets. Cultures that tell more stories about markets tend to tell more stories about prosociality, morality, and stratification and fewer stories about nature. In ethnography as well, markets are associated with prosociality (though not morality), stratification, and anthropocentrism. To assess the direction of causality, I assemble a global dataset of ancient trade routes and hubs, from which I derive a measure of historical trade exposure for each society. Using this measure as an instrument for the text-based market indices, I find that causality runs from markets to culture and social structure. Lastly, I develop a separate ethnographic index for capitalism and show that the social and cultural correlates of capitalism are distinct from those of markets.

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1. Introduction

How do markets shape the way we behave, interact, and understand the world? Debates over the social and cultural consequences of markets and capitalism have a long history (Bowles, 1998; Hirschman, 1982). Some argue that the mutualistic interactions of markets cause an expansion of the moral circle and encourage civic values (Enke, 2023; Harris et al., 2023; Henrich et al., 2010; Montesquieu, 1748; Schilpzand & de Jong, 2023). Others argue that the impersonal, ephemeral, competitive conditions of markets erode moral responsibility and select for selfishness (Falk & Szech, 2013; Graeber, 2014; Marx & Engels, 1906; Polanyi, 1944; Weber, 1921/1978). Markets may have both evolutionary and contextual effects on culture and behavior, influencing what preferences and beliefs persist and spread as well as how they express as behavior (Bowles, 1998). And the evolutionary pathway can be further decomposed into transmission network effects—who interacts with whom—and selection effects—which preferences and beliefs are adaptive in those interactions.[1] Markets may also lead to stratification with social hierarchies arising alongside economic disparities as a means of justifying and thereby stabilizing inequality (Bright et al., 2025; Folbre, 2025; Mason et al., 2022; Zinn, 2015). Such hierarchies might even extend beyond the human world. Whereas many traditional Indigenous knowledge systems assign sacredness and personhood to the more-than-human world, the dominant worldview in modern market societies conceptualizes nature solely as an object for human use (Abram, 2012; Kimmerer, 2013; Leopold, 1949).

Empirically testing hypotheses concerning the social and cultural effects of economic systems is challenging. The historical conditions of societal divergence rarely approximate a natural experiment, and modern market societies (where we have the best available data) represent only a narrow sliver of the diverse range of social, cultural, and economic configurations apparent in archaeological and ethnographic records. One approach to cross-societal cultural economics is to use folklore text as data. This method was pioneered by Michalopoulos and Xue in their paper, "Folklore" (2021), where they transform Berezkin's global catalogue of folklore motifs (2015) into a dataset of the textual prevalence of various concepts in almost a thousand folklore traditions. This large sample represents the substantial diversity across human societies. Michalopoulos and Xue's econometric resuls show that geographical, political, cultural, and economic conditions are

[1] For example, in "Trade and Geography in the Spread of Islam," Michalopoulos, Naghavi, and Prarolo show how trade routes promoted the spread of Islam by exposing more people to the religion and by increasing the incentives of conversion (2018).

reflected in folklore. A number of subsequent papers have employed their data to explore the economic determinants of culture (Becker, 2024; Cao et al., 2021; Enke, 2023; Eruchimovitch et al., 2024; Fan et al., 2024). In "Market Exposure and Human Morality", Enke analyzes the effects of markets on morality and prosociality using indices constructed from Michalopoulos and Xue's folklore data and distance to historic trade routes as an instrument (2023).

In this paper, I extend this method to new questions and new data. I analyze the effects of markets on prosociality, morality, ecocentrism, and stratification using folklore and ethnographic text as data. I construct folklore indices based on Michalopoulos and Xue's folklore data for each societal trait of interest, and I develop a new dataset of indices based on ethnographic text from the electronic Human Relations Area Files (eHRAF) World Cultures database. I also assemble a global map of ancient trade routes and hubs, derive a measure of historical trade exposure from the map, and employ the measure as an instrument to assess the direction of causality.[2] Lastly, I construct an ethnographic index for capitalism to investigate whether the roles, relations, tasks, contexts endemic to capitalism produce different social and cultural forms than markets, money, and trade more broadly.

2. Folklore and ethnographic text as data

Before delving into the methods, it is important to define folklore and consider why folklore and ethnographic text reflects a people's social, cultural, and economic realities. I use the term folklore as Yuri Berezkin does to refer simply to the traditional stories of a people (2015). As the definition suggests, folklore is primarily transmitted vertically from generation to generation within a group. Consequently, folktales spread when a group migrates or proliferates (Berezkin, 2015; Michalopoulos & Xue, 2021). They may also spread via horizontal transmission, passing from one group to another (Berezkin, 2015; Boas, 1916; Michalopoulos & Xue, 2021, Online Appendix). Different stories persist and spread with differential success. Thus, their dispersal might be modelled in an evolutionary manner (Tehrani, 2023; Toelken, 1996; Wilson, 1976). Through the selective transmission of stories, folklore comes to reflect social, cultural, political, economic, and natural conditions. If a story ceases to be relevant, useful, insightful, relatable, or entertaining, it

[2] Enke (2023) includes the global sample of cultures in his IV analysis, but the trade data he uses covers only the Old World. For my IV analysis, I expand the map of ancient trade to the entire world. I also do not implement ecological polarization as an instrument for market exposure as Enke does (2023). I am not convinced that the exclusion restriction holds for this latter specification since ecology may influence culture directly as well as through markets.

may not be passed on, or it may be altered (Bascom, 1953; Dundes, 1969). The fitness (propensity to persist and spread) of specific stories is thus a function of their relevance, usefulness, insightfulness, relatability, and entertainment value as well as their carriers' own capacities to persist and spread themselves and their cultural influence. And folklore can affect the group's fitness by conveying preferences, beliefs, and behavior adapted to their specific human ecology. Stories serve as vessels for education and enculturation by describing the patterns of the natural world, instilling a sense of meaning and sacredness, inculcating and maintaining social and moral norms, and developing a sense of group identity (Akerlof & Snower, 2016; Bascom, 1954; Michalopoulos & Xue, 2021, Online Appendix; Witzel, 2012). Stories may also aim to legitimize the status quo (Akerlof & Snower, 2016; Bascom, 1953; Wrenn, 2021) or challenge it (Marshall, 2012). In short, folklore is both a reflection of society[3] and a societal force.

Ethnography, like folklore, portrays the characteristics of a people and their environment but from a different point of view. Ethnographic studies collect information through participant observation and interviews. While folklore represents an emic (insider) perspective by definition, ethnography is most often written from an etic (outsider) perspective (Dundes, 1969). And whereas folklore allows a glimpse into the worldview of a people—their ontology, their epistemology, their values (Toelken, 1996)—ethnography details the social structure, cultural norms, and behavior in a society. The folklore and ethnographic data in this paper also complement each other in their breadth and depth: Berezkin's catalogue of folklore motifs covers more cultures than the eHRAF World Cultures database, but the latter contains more text per culture. Figure 1 maps out the cultures included in the folklore and ethnographic data.

[3] This is sometimes referred to as reflection theory in sociology.

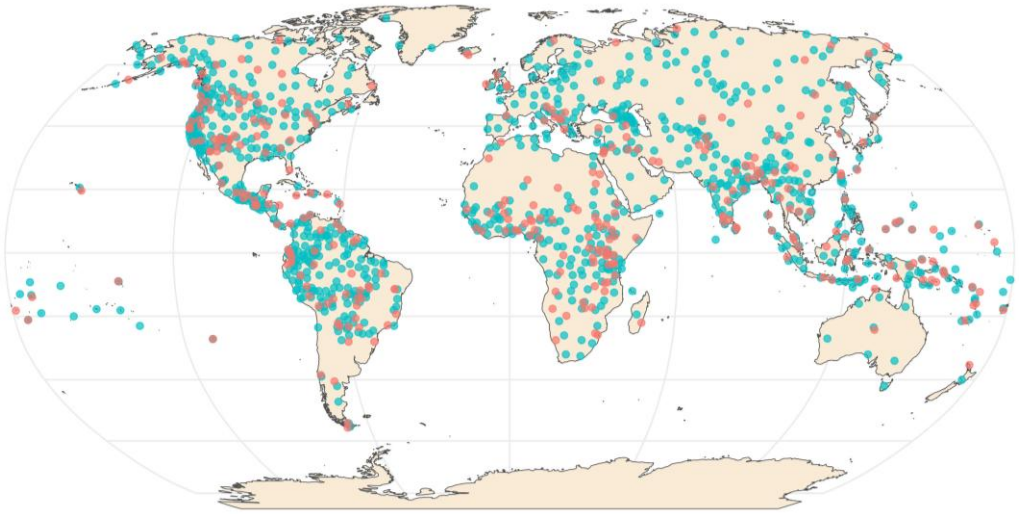


FIGURE 1. MAP OF CULTURES IN FOLKLORE AND ETHNOGRAPHIC DATASETS

Notes: The societies in the folklore and ethnographic datasets span the globe. Turquoise points represent cultures in the folklore data. Orange points represent cultures in the ethnographic data.

2.1. Folklore data

2.1.1. Berezkin's catalogue of folklore motifs (2015)

The folklore text I use comes from a catalogue of 2,564 folklore motifs from 958 folklore traditions that Yuri Berezkin compiled from over 6,000 sources (2015)[4]. A folklore motif, according to Berezkin, is an episode or an image that appears in multiple folklore traditions (2015, p. 37). Anthropologists and folklorists have long noted the redundancy of images and episodes across different folklore traditions, and several efforts have been made to catalogue these motifs and their spatial and cultural distribution, but Berezkin's catalogue is by far the most extensive and globally representative. The source texts include books and articles published mostly in the 20th century, but the folktales themselves may be hundreds, thousands, or even tens of thousands of years old.[5] In the catalogue, each motif has a reference ID, a title, and a one or two sentence description. The contents range from mythological representations of the sun and moon to etiologies of natural phenomena to tales of tricksters. Table 1 describes the catalogue and the distribution of the motifs. Table 2 presents two example motifs from the catalogue and a sample of the folklore traditions in which they appear.

[4] This catalogue first became available in 2015 and has been updated several times, the latest time being in 2022. Currently, it contains 2,945 motifs associated with 999 folklore traditions. I use the Michalopoulos and Xue's dataset which they derive from the 2019 version of the catalogue.

[5] Some folklore motifs in the Berezkin catalogue (such as the Cosmic Hunt) may be of paleolithic origin (d'Huy, 2013).

TABLE 1. SUMMARY OF FOLKLORE DATA

Continent	*Cultures*	*Motifs per culture*	*Sources per culture*	*Publication year*
Africa	137	55.1 (51.3)	12.9 (9.7)	1961.5 (16.4)
Asia	204	92.8 (87.9)	15.2 (13.2)	1957.4 (18.4)
Europe	138	196.0 (134.2)	28.7 (18.1)	1966.4 (15.7)
North America	245	74.3 (47.5)	12.0 (9.8)	1952.0 (20.3)
Oceania	55	41.3 (21.5)	9.9 (5.8)	1944.6 (16.9)
South America	163	58.3 (40.9)	9.8 (8.6)	1966.1 (19.3)
World	943	88.6 (88.1)	14.8 (13.2)	1958.7 (19.5)

Notes: This table summarizes the global distribution of folklore motifs in Berezkin's catalogue, including the number of cultures, the average number of motifs and sources per culture, and the average publication year of the sources. Standard deviations are in parentheses.

2.1.2. Michalopoulos and Xue's folklore concept data (2021)

In their 2021 paper, "Folklore," Michalopoulos and Xue transform Berezkin's catalogue of folklore motifs into a dataset of the prevalence of various concepts in the folklore traditions. This dataset consists of 958 culture observations and 9,005 concept variables. To create the concept variables, they use a semantic network project called ConceptNet to amass a list of 9,005 common words (the 10,000 most common words minus NLTK stop words like "a" and "the"); for each of these "concepts," they obtain a list of the 50 most closely related words according to ConceptNet, and they count the number of motifs in each culture that contain any of these related words in the motif title or description. In short, each society is associated with a number of folklore motifs, and each motif is associated with a number of concepts. Michalopoulos and Xue's concept variables count the number of motifs in each folklore tradition associated with each concept. For example, 16 out of the 152 folklore motifs of the Arapaho people are tagged with words relating to "earth," so the "earth" concept variable has a value of 16 for the Arapaho observation. Table 2 shows a sample of the concepts associated with motifs I11 and M91b.

TABLE 2. FOLKLORE MOTIFS AND CONCEPTS

ID	*Motif title*	*Motif description*	*Cultures*	*Concepts tagged*
I11	Cosmic turtle or toad	A turtle, toad, or frog supports the earth or is its embodiment.	Arapaho Aztec Bengali Bulgarians Buryat Cheyenne Chinese *… and 55 other cultures*	aid aspect assist assistance backer backing behalf *… and 50 other concepts*
M91b	The sold ashes	Using a trick, a man sells or exchanges ashes for treasure. Another person tries to sell ashes and is ridiculed.	Amhara Assamese Bashkir Bengali Buryat Georgians Kazakh *... and 45 other cultures*	accept accomplish accomplishment achievement acquisition administer allegedly *... and 159 other concepts*

Notes: These two folklore motifs from Berezkin's catalogue (2015) exemplify how each of the 2,564 folklore motifs is tied to multiple cultural traditions and multiple concepts (Michalopoulos and Xue, 2021). Michalopoulos and Xue created the concept variables by counting the number of motifs in each culture tagged with the concept (2021).

2.2. Ethnographic data

The ethnographic text comes from the electronic Human Relations Area Files World Cultures database, an online collection of over 770,000 pages of ethnographies describing over 360 cultures. HRAF anthropologists have classified each culture by subsistence type and indexed each paragraph by subject in accordance with the Outline of Cultural Materials (Murdock et al., 2008). There are more than 700 subjects in this classification scheme which cover a vast range of topics from kinship patterns to religious beliefs and from modes of production to technology. The eHRAF search function returns the number of paragraphs containing any combination of OCM subjects and/or keywords in the collected ethnographies of each culture. For example, by searching "subject: ritual," I find that there are 13 paragraphs in the collected Maasai ethnographies that are tagged with the OCM subject, "ritual," and by searching "text: wage OR profit," I find that there are 29 paragraphs in the collected Kazakh ethnographies that contain one of these words. This search feature allows me to extract the textual prevalence of different themes in a manner similar to Michalopoulos and Xue's folklore concept data (2021). Table 3 presents a summary of the eHRAF World Cultures database.

TABLE 3. SUMMARY OF ETHNOGRAPHIC DATA

Continent	*Cultures*	*Pages per culture*	*Documents per culture*	*Publication year*
Africa	70	1920.0 (1752.1)	17.9 (14.2)	1920.5 (12.5)
Asia	76	2362.7 (2272.7)	18.8 (16.4)	1919.0 (15.9)
Europe	24	2536.0 (2084.5)	19.7 (19.2)	1895.3 (86.7)
North America	78	2643.3 (3581.1)	23.6 (30.4)	1904.8 (17.1)
Oceania	29	2156.2 (1826.7)	21.3 (21.8)	1910.1 (17.6)
South America	47	1212.6 (917.4)	11.1 (8.0)	1909.0 (41.5)
World	346	2232.6 (2423.6)	19.4 (20.4)	1913.3 (31.6)

Notes: This table reports the mean number of ethnography pages and documents and the mean birth year of the ethnography authors. Standard deviations are in parentheses. All meta-data were obtained via the HRAF reference materials webpage or through direct communications with HRAF staff.

2.3. Index construction and validation

To measure the prevalence of prosociality, morality, ecocentrism, stratification, markets, capitalism, and subsistence types in the folklore and ethnographic text of each society, I construct indices using Michalopoulos and Xue's concept variables and the eHRAF World Cultures search engine. This method assumes that folklore and ethnography reflect social, cultural, and economic realities in a way that can be measured via the textual prevalence of relevant words and subjects. In other words, people more exposed to markets tell more stories about markets, and their ethnographies contain more references to markets. The first step in the construction of each index is to choose a list of component terms.

For each folklore index, I select a set of concept variables that tag relevant words. The market concepts tag words related to markets, trade, and money; the prosociality concepts tag words related to caring, giving, helping, and sharing; the morality concepts tag words related to judgements of right and wrong behaviors; the ecocentrism concepts tag words related to animals, plants, the earth, and the sky; the stratification concepts tag words related to hierarchy, rank, domination, and servitude. For each ethnographic index, I choose a set of relevant keywords or subjects. The markets index derives from keywords related to markets, trade, and money; the prosociality index from keywords related to reciprocity, helping, and kindness, and selfishness; the morality index from the "ethics" subject; the ecocentrism index from the co-occurrence of subjects related to sacredness and subjects related to culturally specific concepts of the more-than-human world; and the stratification index from the "social stratification" subject. Table A.1 in Appendix A presents a complete list of the component terms for every index.

The indices were derived from the sets of relevant terms according to the following formula:[6]

$$index_i = \ln\left(1 + \frac{relevant\ terms_i}{amount\ of\ text_i}\right) \quad (1)$$

For folklore, the prevalence of relevant terms is the average proportion of motifs tagged by each relevant concept. And for ethnography, the prevalence of relevant terms is the number of paragraphs tagged by any of the relevant keywords or subjects divided by the number of pages of ethnography.

I test the folklore and ethnographic text-based indices for validity using data coded by ethnographers and geospatial data on proximity to ancient trade, nighttime light, and human modification of ecosystems (Appendix A). The results suggest the markets, capitalism, stratification, and subsistence type indices accurately reflect actual societal conditions. The morality indices correlate with the presence of a moralizing god and the degree of Christianization and Islamization.[7] The ecocentrism indices are positively correlated with each other, and each is negatively correlated with human modification of ecosystems, but of these, only the relationship between ecocentrism in ethnography and human modification of ecosystems is statistically significant ($p = 0.05$). Lastly, the prosociality index validation tests fail to find any effects. Interpretations of the main prosociality results must therefore be confined to the folklore and ethnographic text.

More details on the construction and validation of the indices can be found in Appendix A along with example passages of text tagged by the indices and heat maps of their global distribution.

3. Empirical strategy

To analyze the social and cultural effects of markets and capitalism, I apply regression analysis to the folklore and ethnographic indices. The general model specification for both folklore and ethnography regressions is as follows:

$$y_i = \mu + \alpha E_i + \boldsymbol{B'}_i\beta + \varepsilon_i \quad (2)$$

[6] Before obtaining the indices, I drop the 15 folklore observations that are associated with fewer than 5 motifs, and I drop the 16 ethnographic observations that are associated with fewer than 50 pages.

[7] A higher score on the morality indices does not necessarily indicate more moral behavior—only that the text contains more content on what is right and wrong.

where i denotes the cultural group, y is the social/cultural index of interest, μ is the intercept, E is the economic index of interest, and $\boldsymbol{B}$ is the vector of baseline controls. The coefficient of interest, α, indicates the percent change in the textual prevalence of prosociality, morality, ecocentrism, or stratification associated with a 1% increase in the textual prevalence of markets, holding the baseline controls constant.

Following the precedent of Michalopoulos and Xue (2021) and Enke (2023), I cluster the standard errors of the folklore regressions by language family. Since folklore is a linguistic phenomenon, folklore index values are likely correlated with language family. The baseline controls of the folklore regressions include the number of motifs and its square, the number of sources, and the mean publication year of the source. For the ethnography regressions, the baseline controls include the number of ethnographic pages and documents, the mean birth year of the author, the modal gender of the author, the modal region of origin of the author, the mean beginning and end dates of the coverage range of the documents, and the prevalence of the word "the."[8] By controlling for the demographic characteristics of the "average" ethnographer of each cultural group, I try to account for ethnographer bias.

For the main analysis, I estimate this model first using ordinary least squares then using the method of instrumental variables. I also estimate both the folklore and ethnographic regressions with OLS for several additional specifications to test whether the results are robust to the inclusion of regional fixed effects and subsistence type indices.

3.1. Instrumental Variable Analysis

To test whether the social and cultural effects of markets are causal, I apply the method of instrumental variables using a measure of historical trade exposure as the instrument. I derive this measure from a global map of ancient and early medieval trade routes and hubs which I assembled from a number of sources, including Michalopoulos, Naghavi, and Prarolo's map of Old World trade (2018), the catalogue of ancient ports (de Graauw, 2022), the Pleiades gazetteer of ancient places (Elliott et al., 2025), the Mapping Past Societies project at Harvard (McCormick, 2026), the Old World Trade Routes project (OWTRAD) (Ciolek, 2012), the UNESCO World Heritage List, the *Atlas of World History* (O'Brien, 1999), and *The Complete Atlas of World History*, Volume 1

[8] I control for the prevalence of "the" because a small portion of the ethnographic documents are not written in English.

(Haywood, 1997). The resulting global map of ancient and early medieval trade consists of 7,442 sites (Figure B.1).

I calculate the historical trade exposure for each society by summing across the Gaussian-transformed distances to all trade sites (Equation B.1). Figure 2 presents a global heat map of this measure. I discuss the development of the ancient trade dataset and the historical trade exposure measure in more detail in Appendix B. The F-statistics of the first stage regressions (133 for folklore and 23 for ethnography) indicate a strong instrument. For the exclusion restriction to be satisfied, historical trade exposure must be related to the social and cultural characteristics of a society only through the effects of markets. By interpreting the IV coefficients as the causal effects of markets, I thus presume that the social and cultural traits apparent in folklore and ethnography did not influence the development of ancient trade.

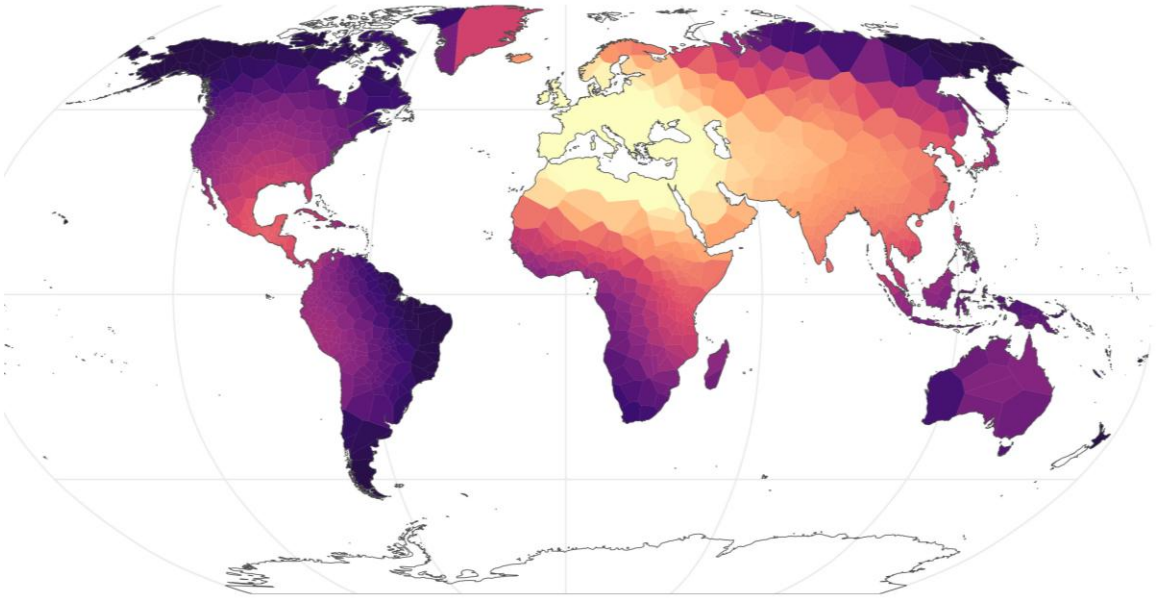


FIGURE 2. MAP OF HISTORICAL TRADE EXPOSURE

Notes: This map plots the historical trade exposure of every society in the folklore and ethnographic datasets. I use a Voronoi tessellation based on societal coordinates to emphasize the geographical variation. The map shows intensive trade activity around the Mediterranean. Trade routes and hubs along the Silk Roads, around the Indian Ocean, and in precolonial Mesoamerica are also apparent.

4. Regression results

Tables 5 and 6 report the regression results from the folklore and ethnographic text-based indices, respectively. In folklore, a 1% increase in markets is associated with a 0.697% increase in prosociality ($p < 0.01$), a 0.674% increase in morality ($p < 0.01$), a 0.257% decrease in ecocentrism ($p < 0.01$), and a 0.177% increase in stratification ($p < 0.01$), holding constant baseline controls (Table 5, Panel A). These coefficients increase in magnitude when estimated using historical trade

exposure as the instrument (Table 5, Panel B). Market exposure thus seems to increase prosociality, morality, anthropocentrism, and stratification in folklore.

In ethnography, a 1% increase in markets is associated with a 0.124% increase in prosociality ($p < 0.01$), a 0.079% decrease in ecocentrism ($p < 0.01$), and a 0.322% increase in stratification ($p < 0.01$), holding constant baseline controls (Table 6, Panel A). The OLS results do not show a relationship between markets and morality in ethnography. The IV estimation yields larger magnitude coefficients than OLS for ecocentrism and stratification, a positive effect on morality ($p = 0.02$), and no effect on prosociality (Table 6, Panel B). Thus, market exposure seems to increase anthropocentrism and stratification in ethnography as in folklore, but the effects on prosociality and morality are less clear.

TABLE 5. THE SOCIAL AND CULTURAL EFFECTS OF MARKETS IN FOLKLORE

	Prosociality	**Morality**	**Ecocentrism**	**Stratification**
		Panel A. OLS		
Markets	0.697***	0.674***	-0.257***	0.177***
	(0.075)	(0.073)	(0.063)	(0.054)
Observations	943	943	943	943
		Panel B. IV		
Markets	1.078***	1.139***	-0.281**	0.448***
	(0.196)	(0.141)	(0.108)	(0.072)
Observations	943	943	943	943

Notes: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors (in parentheses) are clustered by language family. Baseline controls include the number of motifs and its square, the number of sources, and the mean publication year of the source. For the IV regressions in Panel B, exposure to ancient trade is used as an instrument for the text-based market index.

TABLE 6. THE SOCIAL AND CULTURAL EFFECTS OF MARKETS IN ETHNOGRAPHY

	Prosociality	Morality	Ecocentrism	Stratification
		Panel A. OLS		
Markets	0.124***	0.012	-0.079***	0.322***
	(0.032)	(0.010)	(0.013)	(0.057)
Observations	346	346	346	346
		Panel B. IV		
Markets	0.029	0.211**	-0.244***	1.108***
	(0.101)	(0.087)	(0.068)	(0.282)
Observations	324	324	324	324

Notes: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses. Baseline controls include the number of ethnographic pages and documents, the mean birth year of the author, the modal gender of the author, the modal region of origin of the author, the mean beginning and end dates of coverage, and the prevalence of the word "the." For the IV regressions in Panel B, exposure to ancient trade is used as an instrument for the text-based market index. American diaspora cultures (as classified in eHRAF) are dropped.

Figure 3 plots the OLS (a.) and IV (b.) coefficient estimates from regressing the social and cultural indices on the markets index in folklore and in ethnography for three additional specifications. The coefficients from the baseline specifications are plotted in the first rows for comparison. The second and third rows plot coefficient estimates when fixed effects for continent and region[9] are included, respectively. The fourth rows plot coefficient estimates when indices for the textual prevalence of agriculture, pastoralism, and hunting and gathering are included as controls. Isolating intra-continental and intra-regional effects and controlling for subsistence modes tends to slightly diminish the magnitude and statistical significance of the OLS estimates without eliminating them. The IV estimates are less robust to the inclusion of these additional controls. In particular, the coefficient estimates for the effects on ecocentrism become unstable, suggesting inter-regional differences and subsistence types are important moderators of the effects of markets on ecocentrism.

[9] I chose a 23-region classification because further disaggregation results in several subregions containing only one observation.

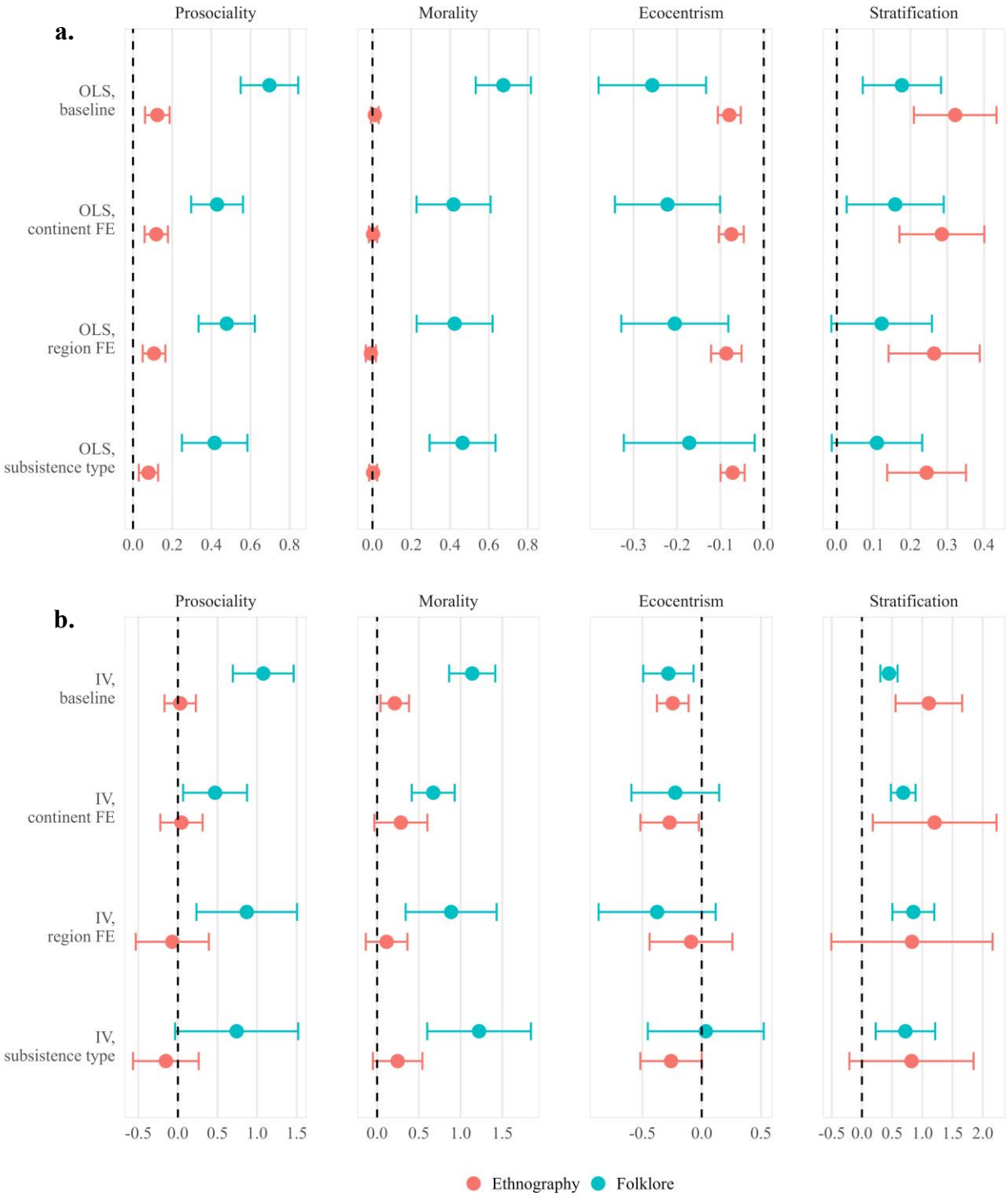


FIGURE 3. ALTERNATIVE REGRESSION SPECIFICATIONS

Notes: These figures report the OLS (a.) and IV (b.) coefficient estimates from regressing prosociality, morality, ecocentrism, and stratification on markets for four different specifications. All regressions include baseline controls. The folklore regressions all have 943 observations, and the ethnography regressions have 346 observations for OLS and 324 for IV. Coefficient estimate error bars represent 95% confidence intervals. The coefficients from the baseline specifications (also reported in Tables 5 and 6) are plotted in the first row of each figure for comparison. The second and third rows plot the coefficients for markets when continent and regional fixed effects are included, respectively. The fourth row plots the coefficients when controlling for text-based indices of agriculture, pastoralism, and hunting and gathering.

5. Interpreting the results

The regression analyses indicate that cultures that tell more stories about markets tend to tell more stories about prosociality, morality, and stratification and fewer stories about nature. In ethnography as well, markets are associated with prosociality, stratification, and anthropocentrism, though not morality. Validation tests and parallel analyses using external data suggest that the textual patterns reflect patterns in the social, cultural, and economic characteristics of the societies themselves (Appendix A). Furthermore, the IV analysis suggests that the causality may flow predominantly from markets to social and cultural characteristics.

After removing the variance associated with the controls, the folklore of southern and central Asia contains the most market-related content while the folklore of the Americas and Oceania contains the least. In the ethnographic dataset, southern Europe has the highest average residual market score while Australia and New Zealand have the lowest. The Christian, Islamic, and Hindu spheres of influence are particularly well-represented at the high end of market exposure, moral content, anthropocentrism, and stratification. On the other end of the distribution of market exposure, Indigenous religions predominate. It is tempting then to conclude that the results observed are nothing more than reflections of the spread of major world religions and their social norms and worldviews along networks of trade (Michalopoulos et al., 2018), but when I control for world religion, the main results persist (Table A.11). So, although the diffusion of world religions along trade routes may be one manifestation of the social and cultural effects of markets, they cannot account for the entirety of these effects.

One potential causal mechanism that could explain the results is cultural evolution (Bowles, 1998; Bowles et al., 2021; Boyd & Richerson, 2005). The incentives and environment of market exchange may exert an evolutionary force on the social and cultural characteristics of a society, influencing which social structures, beliefs, and preferences persist and spread. In this section, I re-contextualize the findings within the relevant literatures and examine each result through the lens of cultural evolution.

5.1. Prosociality and morality

The regression results reveal a strong effect of markets on the prosocial and moral content of folklore (Table 5). In ethnography, the effect on prosociality is weaker, and there is no effect on moral content (Table 6). The null results of the prosociality index validation tests (Table A.5) and

the analysis of external ethnographic data (Table A.10) limit what we can infer about the effect of market exposure on prosociality beyond the text. The morality validation tests show that the indices correlate with the salience of moral themes in religion (Table A.6). In the case of both indices, the relationship between folklore content and behavior is unclear. Does a society with an abundance of desirable behavior tell more or fewer stories about desirable behavior? The indefinite validation results may be partly attributable to this ambiguity.

In the experimental literature, several cross-cultural and inter-group field studies have found a positive correlation between market exposure and prosociality in small-scale societies (Henrich et al., 2004, 2010; Rustagi, 2024; Rustagi et al., 2010). However, experiments varying context rather than culture have often found that market-related frames and environments undermine prosociality and moral consideration (Bartling et al., 2015; Ellingsen et al., 2012; Falk & Szech, 2013; Hoffman et al., 1994; Liberman et al., 2004; Pillutla & Chen, 1999; Roth et al., 1991). One way of reconciling these apparently contradictory findings is by differentiating between the effects of markets on what preferences persist and spread—i.e. cultural evolution—and the contextual effects of markets on the behavioral expression of preferences. Anonymity, ephemerality, and competition in the context of markets may undermine mechanisms such as empathy and reciprocity that evolved to promote prosociality in socially embedded environments (Garrison, 2026; Schmid et al., 2021), but groups that maintain enough baseline civility among strangers for market exchange to occur might reap a fitness advantage based on the gains from trade between socially distant peoples. In other words, markets create an incentive to extend the moral circle. This does not imply that prosocial and moral behavior is more prevalent overall in market societies but rather that these societies may develop substitutes for socially contingent mechanisms of cooperation as economic interactions become increasingly disembedded from personal networks.

The positive effect of societal market exposure on the prosocial and moral content of folklore may then reflect an explicit effort to inculcate moral norms for situations where our innate social preferences fail to promote prosocial behavior. This interpretation aligns with the argument that omniscient, moralizing, punitive gods arose as a means of promoting anonymous large-scale cooperation in complex society (Henrich et al., 2010; Purzycki et al., 2016). Thus, different frequencies of prosocial and moral themes in folklore may reflect different functional roles of folklore in society.

5.2. Ecocentrism

The results from both folklore and ethnographic regressions suggest that markets diminish ecocentrism. With increasing specialization and division of labor, much of society comes to rely on markets (human interactions) rather than subsistence activities (human-ecosystem interactions) for their basic needs. Thus, the salience of the human world increases relative to the more-than-human world. This shift might also be accompanied by a similar shift what is considered sacred. The regression results might be interpreted as a reflection of this dynamic. As we depend more on trade with other humans for our subsistence and less on direct interactions with the more-than-human world, we tend to tell fewer stories about the more-than-human world and assign less sacredness to nature.

A similar pattern can be found in creation myths, which exemplify a people's identity and relation to nature (Kimmerer, 2013). In Genesis, a creation story from one of the cradles of complex society, God creates humans in his own image and grants them dominion over the earth and all non-human animals. This cosmogony formulates a hierarchy of nature in which humans hold a privileged and separate status (White, 1967). In comparison, the mythologies of many Indigenous peoples grant personhood to the more-than-human world and conceptualize the relationship between humans and nature in terms of kinship, embedding humans not in a hierarchy of nature but in a web of coinciding material, social, and sacred relations, where humans are tasked not with dominion but with reciprocity (Kimmerer, 2013).

The ontologies of nature apparent in mythology may manifest behaviorally in human-environment interactions (Campbell & Moyers, 2011; Kimmerer, 2013). The validation tests in Appendix A reveal a weak negative correlation between textual ecocentrism and human modification of ecosystems (Table A.7). Similarly, the literature on the worldviews and ecological practices of Indigenous peoples demonstrates the relationship between ecocentric ontologies and sustainable behavior (Brondízio et al., 2021; Garnett et al., 2018).

The behavioral link between markets and ecological practice is evidenced by the association between historical trade exposure and environmental degradation (Table A.10). This effect of markets may operate via two causal pathways. First, markets increase incentives for resource extraction by providing outlets for surplus. Second, markets may erode ecocentric preferences by distancing actors from the environmental costs of their actions and thereby

weakening the fitness advantage of sustainable practices. This could be a fruitful direction for future theoretical research.

5.3. Stratification

The effect of markets on stratification in folklore and ethnographic text is strong and robust (Tables 5, 6, and A.10). As the validation tests reveal (Table A.8), the text-based indices for stratification correlate with gender bias in folklore (Michalopoulos & Xue, 2021) and ethnographic measures of class differentiation, caste differentiation, and slavery. Thus, the textual prevalence of stratification reflects social, cultural, and economic forms of stratification in society. By economic stratification, I refer to disparities in economic opportunities and outcomes. By social stratification, I refer to asymmetries of power and privilege in the social and political organization of society. By cultural stratification, I refer to socially learned and constructed concepts of otherness and hierarchical classification schemes.

Markets may generate economic inequality. Then, social and political hierarchies may arise from and intensify this economic inequality (Gintis, 2016, Chapter 2). Myths describing the system as natural and the outcomes as deserved may be used to justify social and economic stratification (Akerlof & Snower, 2016; Wrenn, 2021). While the selective diffusion of cultural traits (such as myths) is often thought of as a decentralized, unconscious process driven by random "mutation" and voluntary emulation, the dissemination of myths and ideologies of hierarchy and othering seems more likely to be an intentional project of the privileged to justify their position.

A popular historical theory attributes stratification in complex society to the invention of agriculture (e.g. Diamond, 1999; Harari, 2015; Kaplan et al., 2009). Agriculture allows for the creation and storage of surplus; individuals with more surplus then employ their wealth to expropriate surplus from others. Simultaneously, the existence of surplus promotes markets and trade (as a vent for the surplus). One might argue then that the relationship between markets and stratification in folklore and ethnographic text is reflecting the development of agriculture. However, as Figure 3 shows, controlling for subsistence type does not eliminate the effect of markets on stratification. At least some of the effects of markets are separable from those of agriculture.[10]

[10] David Graeber suggests that the rise of coinage in the Axial Age is intimately tied to slavery through war: coins were needed to pay soldiers on far-ranging military expeditions, and to meet this demand for coins, war captives were forced to mine for gold and silver as slaves (2014).

6. Markets versus capitalism

It is not immediately clear whether the social and cultural effects of markets observed in folklore and ethnographic text also pertain to the economic system of capitalism. Markets and trade can be found across much of the ethnographic and historical records, but capitalism is a historically specific economic system where markets take on a distinct form and role in society (Polanyi, 1944). Capitalism is characterized by private ownership of the means of production and the wage-labor relation (Wood, 2002). In capitalism, markets are essential for social reproduction—i.e. most people in a capitalist society must sell their labor for a wage to provide for their own subsistence (Wood, 2002).

To tease apart the social and cultural effects of capitalism from those of markets, I develop an index for capitalism based on the textual prevalence of terms related to capitalism, employment, wages, and profits in ethnographies (Table A.1). I do not construct a corresponding index from folklore text because none of the motifs in Berezkin's catalogue relate specifically to capitalism. I then regress each of the social and cultural indices on the indices for capitalism and markets along with the baseline controls. The eHRAF subsistence type classification includes a category for commercial economies, so I also regress each of the social and cultural indices on this indicator variable. Table 7 reports the OLS coefficient estimates.

A 1% increase in the textual prevalence of capitalism is associated with a 0.135% decrease in the prevalence of prosociality ($p = 0.04$), a 0.114% decrease in the prevalence of ecocentrism ($p < 0.01$), and a 0.626% increase in the prevalence of stratification ($p < 0.01$), holding constant the market index as well as baseline controls (Table 7, Panel A). The market index coefficients are similar in sign and significance to the results when the capitalism index is omitted. These results show distinct effects of markets and capitalism. In particular, prosociality is associated positively with markets but negatively with capitalism in ethnographic text. The results from the eHRAF subsistence type indicator regressions align with the text-based capitalism index results. The ethnographic text of societies classified as commercial economies contains on average 0.027% less prosociality-related content ($p < 0.01$), 0.021% less ecocentrism-related content ($p < 0.01$), and 0.080% more stratification-related content ($p < 0.01$), holding constant baseline controls (Table 7, Panel B). I do not find any relationship between capitalism and the prevalence of moral content in ethnographic text.

TABLE 7. THE SOCIAL AND CULTURAL CORRELATES OF CAPITALISM IN ETHNOGRAPHY

	Prosociality	Morality	Ecocentrism	Stratification
		Panel A. Text-based indices		
Capitalism	-0.135**	0.030	-0.114***	0.626***
	(0.067)	(0.023)	(0.030)	(0.143)
Markets	0.152***	0.006	-0.055***	0.190***
	(0.032)	(0.010)	(0.014)	(0.059)
Observations	346	346	346	346
		Panel B. eHRAF subsistence type indicator		
Commercial economy	-0.027***	-0.004	-0.021***	0.080***
	(0.006)	(0.003)	(0.003)	(0.021)
Observations	346	346	346	346

Notes: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses. Panel A reports the results from regressing the text-based social and cultural indices on the text-based indices for capitalism and markets. Panel B reports the results from regressing the text-based social and cultural indices on the commercial economy subsistence type category coded by eHRAF anthropologists. Baseline controls include the number of ethnographic pages and documents, the mean birth year of the author, the modal gender of the author, the modal region of origin of the author, the mean beginning and end dates of coverage, and the prevalence of the word "the."

These results demonstrate that the effects of markets cannot always be generalized to capitalism. The opposite signs of the prosociality coefficients for capitalism and markets suggest that the social and cultural effects of markets depend on the dominant form and role of markets in society. It could simultaneously be the case that local markets and trade create mutualistic relationships that undermine intergroup antagonism (in accordance with Montesquieu's "doux commerce" theory (1748)) and that the market imperative of capitalism induces cut-throat competition that selects for self-interest. In contrast, the negative association between capitalism and ecocentrism may be an intensified version of the market effect. The long supply chains of global capitalism distances individuals from the ecosystems that support them and thus from the environmental costs of their actions even more than pre-capitalist markets. And the profit imperative central to capitalism amplifies the incentives to extract resources and increase production regardless of external environmental consequences and planetary boundaries. Lastly, capitalism is strongly associated with stratification. The relationship between capitalism and economic stratification might be explained by the accumulation, concentration, and centralization of capital and the impoverishment of the proletariat (Marx, 1889, 1932; Sweezy, 1972). As for social and cultural stratification, patriarchal and racial systems of oppression have been instrumental to the rise and development of capitalism (Robinson, 2000; Williams, 1994) and are in turn shaped by capitalism to support its functioning and stability (Bright et al., 2025; Folbre, 2025; Mason et al., 2022).

7. Conclusion

In this paper, I used folklore and ethnographic text as data to analyze the effects of markets on prosociality, morality, ecocentrism, and stratification. The results suggest that cultures more exposed to markets tell more stories about prosociality, morality, and stratification and fewer stories about nature. The relationships between markets and anthropocentrism and stratification appear in ethnographic text as well. An IV analysis using exposure to ancient trade as the instrument suggests that these effects of markets are causal.

Several limitations of the method are worth noting. First, the transformation of the text into data involves decontextualization which could reduce the accuracy and precision of the indices. Validation tests partially mitigate this concern. However, the weak validation results for prosociality and ecocentrism represent a limitation of their own and lower the confidence with which we can generalize the findings beyond the text. Parallel analyses of the external data used for validation and experimental studies asking similar questions address this second limitation by providing non-textual evidence for the social and cultural effects of markets observed in folklore and ethnographic text. Third, the lack of timestamp on the collections of folklore and ethnography associated with each culture limits the sharpness of the indices and muddies the historical interpretation of the results. The dynamic, evolutionary nature of folklore motifs precludes any attempt to assign dates of origin to the motifs, and the ethnographic collections often cover a substantial period of the group's history. Case studies of the historical economic transitions of particular societies could clarify this temporal ambiguity and, more broadly, test the validity of inferring the evolutionary paths of individual societies from cross-cultural analysis.

The analysis of folklore and ethnographic text presented in this paper implies several avenues for future research. First, little theoretical work has been done on the relationship between markets and anthropocentrism. An evolutionary game theoretic model with an ecological component might operationalize the theoretical hypothesis presented in Section 5.2. Second, further work might explore the relationship between text-based indices of cultural traits and modern-day attitudes. This would involve matching responses in modern surveys to groups in the textual data or aggregating measures to the country level (see Michalopoulos and Xue, 2021). One challenge of this vein of research is in interpreting weak or null results, which might be attributable to either invalid indices or cultural change. Third, large language models might be employed as an alternative way of transforming folklore and ethnographic text into data (e.g. Michalopoulos,

2025). This technique might better preserve context-dependent meaning at the cost of a more opaque transformation process. Fourth, the data and methods could be extended to other relationships such as the cultural consequences of agriculture (Alesina et al., 2013; Fan et al., 2024; Talhelm et al., 2014) and pastoralism (Cao et al., 2021). As demonstrated by this analysis and the growing literature it contributes to, collections of folklore and ethnography offer rich textual data for studying the relationships among the social, cultural, and economic characteristics of human society.

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Appendix A. Index construction and validation

In this appendix, I describe the construction of the text-based indices, report the constituent terms (Table A.1) and map the geographical distribution (Table A.2) of each index, test each index for validation against external cross-cultural data, and conduct several robustness checks of the main results using the same external data.

A.1. Index construction

The folklore and ethnographic indices are derived from a list of relevant terms and themes according to Equation 1 in Section 2.3. The constituent terms for each index are reported in Table A.1. In Table A.2, for each index, I plot its global distribution as a heat map, and I present example motifs or ethnographic passages that contain its constituent terms.

To construct indices from folklore, I select sets of concepts from Michalopoulos and Xue's data based on the related words tagged and the context in which they appear in the motifs (Table A.1). In this selection process, I attempted to maximize coverage of relevant words, minimize inclusion of irrelevant words, and minimize redundancy of words (to reduce the number of times

the same motif is counted). I then obtain each index from the log mean prevalence of the relevant concepts in the set of motifs associated with each folklore tradition (Equation 1).

I experimented with other methods of index construction including using the first principal component of the terms (as Michalopoulos and Xue do in their analysis of moralizing gods (2021, pp. 2013-2015)) and indicator variables based on the presence rather than the prevalence of each relevant concept (as Enke does (2023)). The PCA method often yields constituent term weightings with inconsistent signs unless applied to the indicator versions of the concept variables. The PCA indicator indices yield very similar main results but are less comparable to the ethnographic index results due to their different method of construction. I therefore employ the log prevalence version of the folklore indices in this analysis.

As with the folklore index construction, to create the ethnographic indices, I first select a list of relevant terms (Table A.1). This can be done using OCM subjects and/or keywords. The advantage of using keywords is that there are no constraints on my choice of terms, allowing me to derive indices based on the highly relevant terms. The advantage of using subjects is that they capture more of the context of the terms since the paragraphs are tagged by ethnographers who read the texts. One issue with the keyword approach is that a small portion of the ethnographic documents are not written in English. To account for this, I include the prevalence of the word "the" in all regressions. My objective in keyword/subject selection is to choose terms that correspond one-to-one with the societal characteristic of interest. I use the subject approach for the morality and stratification indices because the OCM subjects "ethics" and "social stratification" correspond so well with the indices, and I use the subject approach for the ecocentrism index to identify paragraphs containing both culturally specific concepts of the more-than-human world and cultural concepts of sacredness, myths, and spirituality. For all other indices, I use the keyword approach. I then obtain the indices from the log prevalence of relevant terms (Equation 1).

TABLE A.1. TEXT-BASED INDEX COMPOSITION

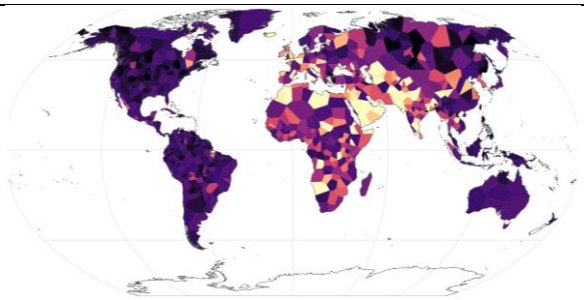
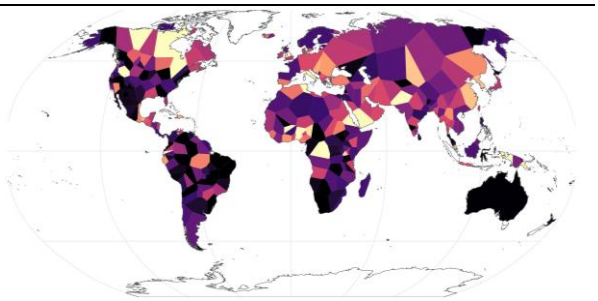
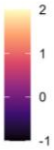
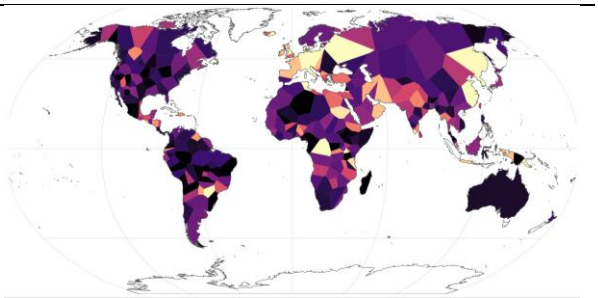
	Folklore		*Ethnography*
Index	Concepts	Related words	OCM Subjects/*Keywords*
Agriculture	• farmer • harvest • plow • rice • sow	• agricultural • agriculture • cereal • corn • crop • cultivate • farm • farmhand • fertilize • furrow • grain • harvest • haymaking • maize • peasant • pig • plough • plow • ploughman • plowman • scatter • seed • sow • sprout • wheat	• *agricultural* • *agriculture* • *crops* • *cultivate* • *cultivating* • *cultivation* • *farm* • *farmer* • *farming* • *garden* • *gardening* • *harvest* • *harvesting* • *horticultural* • *horticulture* • *planting* • *sow* • *sowing* • *sown*
Capitalism	N/A	N/A	• *capitalism* • *capitalist* • *corporation* • *employee* • *employer* • *employment* • *profit* • *wage*
Ecocentrism	• animal • buffalo • crow • fish • flower • forest • fox • grass • hill • insect • moon • ocean • plant • rabbit • rain • rat • reptile • river • season • sky • snow • soil • spider • star • sun • tree • turtle • wind • wolf	• arachne • animal • amphibian • beast • beetle • bird • bison • bloom • blossom • blizzard • brook • buffalo • bunny • buzzard • celestial • cliff • coniferous • crocodile • crow • cub • dam • deer • dirt • downriver • downhill • earth • elk • episode • era • finale • firmament • fish • flower • fox • grass • ground • hare • hawk • heaven • heavenly • heliacal • heron • hill • horizon • hummock • insect • land • lion • lizard • lunar • magpie • mammal • mizar • moon • mosquito • mountain • mouse • moth • ocean • oceanic • orbit • owl • pet • plant • precipitation • rabbit • rain • rainwater • rainy • rat • raven • reptile • reptilian • ridge • river • rooster • season • sea • seawater • serpent • shell • sky • snail • snake • snow • snowfall • soil • solar • sparrow • spider • spiderman • slope • spring • star • starman • stellar • stream • summer • sun • sunlight • sunny • sunrise • sunset • terrapin • thunderstorm • tortoise • tree • treelike • turtle • wasp • watercourse • weather • web • werewolf • wind • winter • wolf • year	Any of the following: • animism • mythology • sacred objects and places • spirits and gods together with any of the following: • ethnobotany • ethnogeography • ethnometeorology • ethnozoology
Hunting / gathering	• fisherman • hunter	• chase • chaser • deer • fish • fisher • fisherman • hunt • hunter • scavenger • stalker	• *fish* • *fisher* • *fishing* • *forage* • *forager* • *foraging* • *gatherer* • *hunt* • *hunter* • *hunter-gatherer* • *hunting*

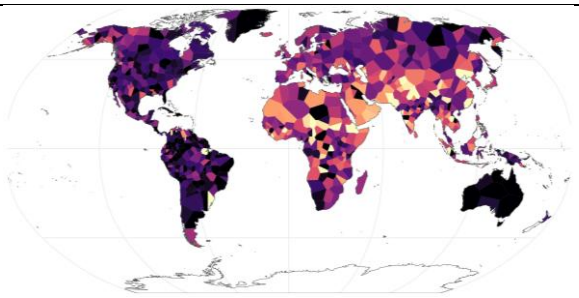
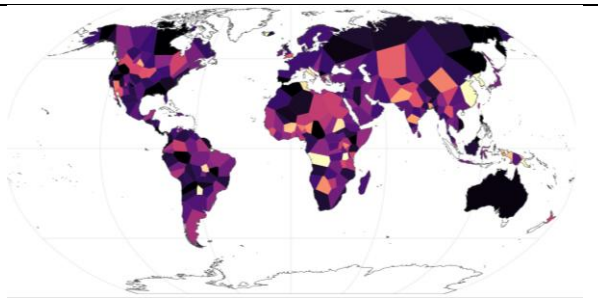
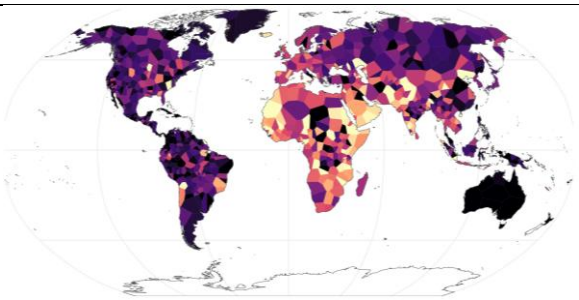
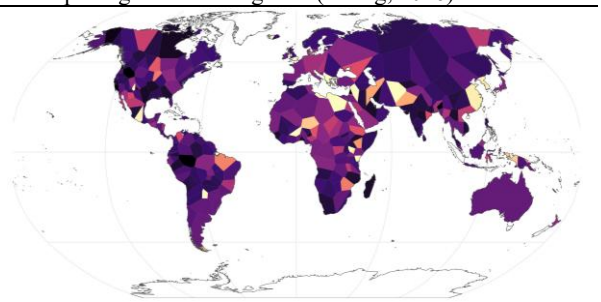
Markets	• cheap • income • market • money • seller • trade	• barter • bazaar • buy • bargain • buyer • cheap • cheaply	• cost • exchange • expensive • fund • income • market • merchant	• money • price • sale • sell • seller • swap • trader	• *bazaar* • *bought* • *buy* • *buyer* • *buying* • *coins* • *currency* • *exchange* • *exchanging* • *market* • *marketplace*	• *merchant* • *money* • *price* • *sale* • *sell* • *seller* • *selling* • *sold* • *trade* • *trader* • *trading*
Morality	• deceive • deserve • duty • evil • forbid • forgive • guilty • injustice • justify • offend • ought • proper • punish • respect • rule • shame • sin • virtue • wrong • wrongdoing	• accuse • admire • annoy • appropriate • appropriately • ashamed • astray • award • ban • burden • cajole • cannot • chastity • cheat • confuse • convict • convince • correct • corrected • correctly • corruption • culprit • deceit • deceitful • deceiver • deception • decent • dishonor • disgrace • disappoint • disturb • duty • dupe • earn • endanger • esteem • error • evil • excuse • explain • forbid	• forgive • fraud • get • gratitude • grief • guilty • guideline • harm • honesty • honor • hoodwink • humiliation • hurt • improper • impede • inappropriate • injustice • injure • insult • inequality • incorrectly • justification • justify • law • mercy • mistake • mistakenly • molest • must • necessary • need • negligence • obligation • offend • ought • patience • person • pity • precise • pretext • prevent	• prohibit • prohibition • proper • properly • rationale • receive • regard • reign • repent • repentance • responsibility • respect • right • ruler • rule • scold • seduce • shame • shameful • sin • sinner • sinful • suitable • sympathy • task • threat • tormentor • torture • trick • unjust • unjustified • unworthy • violate • virtue • want • wife • worthy • wrong • wrongly	• ethics	
Pastoralism	• graze • herd • mule • pastoral	• camel • cattle • cow • donkey • flock • foal • graze	• herd • herder • herdsman • hoof • horse • livestock • mule	• nibble • ox • pasture • saddle • sheep • shepherd • swineherd	• *graze* • *grazing* • *herd* • *herder* • *herding* • *husbandry*	• *livestock* • *pastoral* • *pastoralism* • *pastoralist* • *pasture*
Prosociality	• caring • generosity • gift • given • help • sharing • sympathetic	• affectionate • aid • assistance • benevolent • care • caregiver • compassionate • contribute • donate	• generous • generously • gift • give • giver • goodwill • grant • gratitude • help	• helpful • lend • present • relief • rescue • share • support • sympathize • sympathy	The log prevalence of • *altruism* • *altruistic* • *empathy* • *generosity* • *generous* • *gift* • *helpful* minus the log prevalence of	• *kindness* • *reciprocal* • *reciprocate* • *reciprocity* • *sharing* • *sympathy*

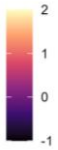
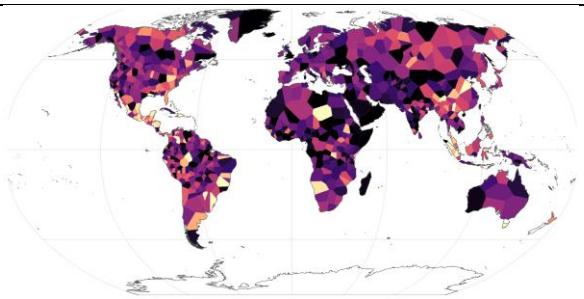
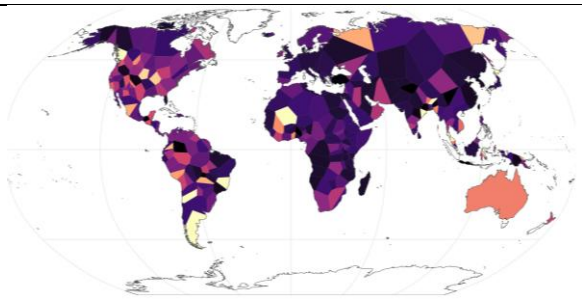
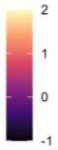
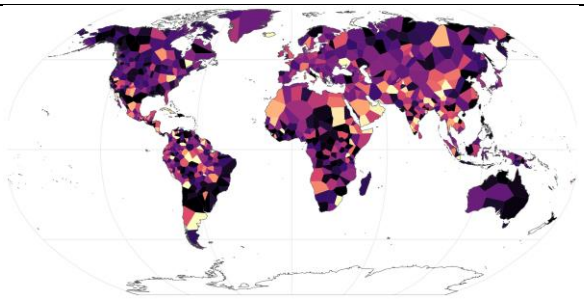
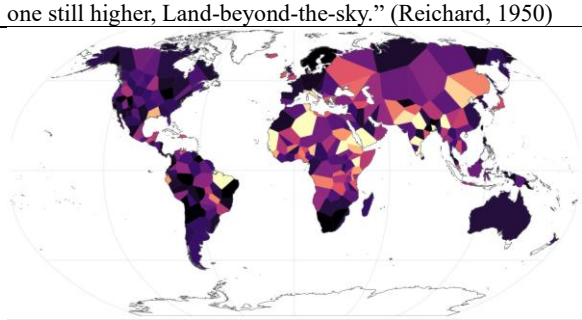
		• generosity	• helper		• *antagonistic* • *antisocial* • *greed*	• *greedy* • *selfish* • *selfishness*
Stratification	• dominate • obedience • rank • servant	• comply • conquer • control • disobedience • dominate • domination • hierarchy	• housekeeper • lord • maid • maidservant • obedient • overtaken • rank	• seniority • servant • slave • status • subservient • supremacy • tier	• social stratification ◦ age stratification ◦ gender status ◦ ethnic stratification ◦ castes ◦ classes ◦ serfdom and peonage ◦ slavery	

Notes: All folklore indices are derived from Michalopoulos and Xue's concept variables (2021), which tag motifs that contain words related to the concepts. Both concepts and related words are reported in the table. Ethnographic indices are constructed either from the paragraph-level tagging of OCM subjects (underlined) or from the keywords that appear in the text itself (in italics).

TABLE A.2. GLOBAL VARIATION AND EXAMPLE TEXT FOR MAIN INDICES

Index	*Folklore*	*Ethnography*
Markets 	 m91b1: The sold skin "A man goes to sell a skin of domestic animal and on his way, by trick or thanks to chance, gets a big sum of money. Usually coming back he explains that this was the price of the skin but when other people kill their animals they cannot sell skins for such a sum."	 "Almost every day some woman sets off with a basket or two of produce to sell in the markets of Bangkok. People buy a sewing machine to earn money by making clothing to order, or a portable gasoline motor that can be rented for cash up and down the canal." (Hanks, 1972)
Capitalism 		 "Employment of wage labour is considered to be the main index of exploitative income: first, because it seems to be the most important and widespread source of expropriated labour among Nupe peasants; and second, because it is under the prevailing conditions the most direct and characteristic criterion of the rural capitalist development which is dawning at present in Nupeland." (Kohnert, 1979)

Prosociality 	 k151: The fisherman and his wife “Supernatural creature fulfills a poor man’s moderate request. After this he or his wife asks for ever bigger gifts till the angry helper punishes them (usually takes all his gifts away).”	 “Instrumental wanglai and expressive wanglai are similar to instrumental exchange and expressive exchange (Befu, 1966–7, 1968, 1974, 1977), and Yan's (1996) ‘instrumental gift giving’ and ‘expressive gift giving’. However Yan also uses categories of ‘negative reciprocity’, ‘unbalanced reciprocity’, ‘obligatory gift giving’, ‘unilateral gift giving’, ‘imbalanced gift exchange’, and fails to differentiate between different specific types of gift giving, as well as combining typologies of exchange and reciprocity, all of which this author finds confusing and complicated. This model, therefore, simply keeps Sahlins's (1972) ‘negative reciprocity’ as negative wanglai , divides his ‘balanced reciprocity’ into instrumental and expressive wanglai , and renames his ‘general reciprocity’ as the more descriptive generous wanglai .” (Chang, 2016)
Morality 	 l106a: Stomach cut open “An antagonist makes a demand to the hero which is correct in form but really is unjustified. The hero fulfills the claims or is punished. Now antagonist takes an object or animal possessed by the hero, is unable to give it back and is punished.”	 “In actuality, there is no rigid separation of esoteric and exoteric knowledge. Probably an intelligent layman among the Navahos picks up a good deal of esoteric knowledge during his lifetime. On the other hand, the fact that such knowledge is regarded as esoteric means that it is not cited in public moral discourses or in the teaching of children. The essential beliefs involved in ethical discourse are therefore entirely exoteric for the average Navaho. Ethics can be talked about freely.” (Ladd, 1957)

Ecocentrism 	 A32: Figure on lunar disc "On the lunar disk visible figure or imprint of a creature or object. (For statistical calculations, this motif also includes the motives A32A - A32J: a rabbit, a frog, a beast of prey, a man, a tree, etc., are visible on the moon.)"	 "Since for eons man has been advancing toward oneness with the universe, he identifies himself with all its parts. This world may be considered a functioning central world; others, left behind but remembered in myth, are underneath; there are others above. The number of worlds is hypothetical, there being little agreement about it; myth furnishes details of four underworlds, of the sky immediately above, and of one still higher, Land-beyond-the-sky." (Reichard, 1950)
Stratification 	 k67b: Bargain not to become angry "Person of a low social position (a man) makes an agreement with a person of high social position (an ogre) that the master must never become angry with the servant. The servant abuses the master until the latter erupts in anger and has to be severely punished or to pay a great fee."	 "In a sense the caste structure constituted the basis of traditional society. Tanjore District in particular has been known for the rigidity and complexity of its caste structure. In the village this structure not only divided the population into sections of unequal ritual status, but also dominated economic and political life. The fundamental importance of the caste structure to the social life of the village can be seen in its settlement pattern (chap. ii), which clearly segregates the three primary segments—Brahmins, Non-Brahmins, and Adi-Dravidas—from one another." (Béteille, 1971)

Notes: This table presents Voronoi heat maps and textual examples for every index used in the main analysis. The plots are based on standardized residuals obtained by regressing the indices on the baseline controls. The units in the legend are in standard deviations. I exclude non-Indigenous American groups from the ethnographic maps so that they are comparable with the folklore maps (Berezkin's catalogue includes no non-Indigenous American groups). Example passages were selected based on the number of index component terms tagged. All folklore motifs come from Berezkin's catalogue (2015), and all ethnography paragraphs were retrieved through eHRAF World Cultures.

A.2. Index validation

To test the indices for validity, I assembled a cross-cultural concordance with the help of the concordance resources of eHRAF and the Database of Places, Language, Culture and Environment (D-PLACE) as well as the concordance developed by Michalopoulos and Xue (2021). 288 cultural observations are present in both the folklore and ethnographic data. The folklore and ethnography samples also overlap with the Ethnographic Atlas (608 and 287 cultures, respectively), the Standard Cross-Cultural Sample (182 and 184), and Carneiro's dataset (113 and 71) (Carneiro, 2025; Murdock et al., 2025; Murdock & White, 2026). I retrieved ethnographic variables from

these datasets on D-PLACE related to prosociality, morality, stratification, markets, and subsistence type (Kirby et al., 2016). For each of the text-based indices, I find the residuals after removing variation associated with the baseline controls. I compare the folklore and ethnographic indices to each other and to the ethnographic variables from the EA, SCCS, and Carneiro datasets. Modern data on nighttime light and human modification of ecosystems is also employed to validate the capitalism and ecocentrism indices, respectively, but these regressions are estimated for only cultures in Africa, Asia, and Europe. Due to settler colonialism in the Americas and Australia, much of the human activity apparent in these regions is attributable to peoples other than those represented in the folklore and ethnographic data.

All folklore indices are positively correlated with their ethnographic counterparts, but the p values for the prosociality, ecocentrism, and stratification indices exceed the conventional 0.1 threshold required for statistical significance ($p = 0.17$, $p = 0.31$, and $p = 0.21$, respectively). Both market indices are strongly correlated with log exposure to ancient trade and proximity to nearest trade site (Appendix B) and measures of exposure to markets, trade, and money in SCCS and Carneiro samples (Barry & Schlegel, 1982; Carneiro, 2025; Murdock & Morrow, 1970; Murdock & Provost, 1973) (Table A.3). The capitalism index is strongly correlated with the commercial economy indicator in eHRAF, log exposure to ancient trade, proximity to nearest trade site, and log nighttime light in the Old World (Elvidge et al., 2021) (Table A.4). The prosociality indices are unrelated to the cultural emphasis on teaching generosity to children (Barry et al., 1976) and to prosocial behavior in experimental games (Henrich et al., 2004, 2010) (Table A.5). The folklore index for morality is correlated with degree of Christianization or Islamization (Rosenblatt et al., 2011), and both the folklore and ethnographic morality indices are correlated with the presence of a moralizing god (Murdock et al., 2025) (Table A.6). Both ecocentrism indices are negatively correlated with the degree of human modification of ecosystems in the Old World (Theobald et al., 2025), but only the ethnographic index effect is statistically significant (Table A.7). The stratification indices are correlated with gender bias in folklore (Michalopoulos & Xue, 2021) and measures of class and caste differentiation and slavery in the EA, SCCS, and Carneiro samples (Carneiro, 2025; Murdock et al., 2025; Murdock & Provost, 1973), but the relationships between the folklore index and SCCS and Carneiro measures are statistically insignificant (Table A.8). The folklore and ethnographic indices for agriculture, pastoralism, and hunting and gathering are correlated with the eHRAF subsistence type categories and with measures of subsistence type in

the EA, SCCS, and Carneiro samples (Barry & Schlegel, 1982; Carneiro, 2025; Murdock et al., 2025) (Table A.9).

Michalopoulos and Xue (2021) and Enke (2023) compare their cultural indices to modern survey data on values, but since many of the societies in the data have experienced significant social, cultural, and economic change since the historical period reflected in folklore and ethnography, a null result using this technique might indicate either an invalid index or cultural change. Additionally, most survey data can only be compared to the folklore and ethnographic text at the country level. Country-level comparisons lose a substantial amount of variation due to aggregation, and in many cases, migration has substantially changed the demographic composition of the country, invalidating the implicit assumption of regional continuity in these comparisons. This is not to say that the text-based indices can never be compared to the modern survey data on values, but such an analysis is beyond the scope of this paper.

TABLE A.3. MARKET INDEX VALIDATION RESULTS

	Ethnographic market index	**Log exposure to ancient trade**	**Log distance to ancient trade**	**SCCS markets**	**Carneiro markets**
			Panel A. Folklore		
Folklore market index	0.155***	0.996***	-0.544***	0.099***	0.114***
	(0.045)	(0.132)	(0.093)	(0.018)	(0.030)
Observations	288	943	943	182	113
			Panel B. Ethnography		
Ethnographic market index		0.472***	-0.291***	0.093***	0.119***
		(0.121)	(0.091)	(0.014)	(0.033)
Observations		324	324	184	71

Notes: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses. Each external measure of market exposure is regressed on the standardized residuals of the text-based market index (obtained by regressing the index on the baseline controls used in the main analysis). Panel A reports the coefficient estimates for the folklore index, and Panel B reports the coefficient estimates for the ethnography index. The measure of exposure and distance to ancient trade are derived from the global dataset of ancient trade routes and hubs described in in Appendix B. The SCCS market exposure index is a derived from the variables SCCS1 (Intercommunity trade as food source), SCCS155 (Money), and SCCS819 (Percent importance of trade in contribution to subsistence and trade) (Barry & Schlegel, 1982; Murdock & Morrow, 1970; Murdock & Provost, 1973). The Carneiro market exposure index is derived from the variables Carneiro4_020 (Regular food surplus production), Carneiro4_091 (Craft production for economic exchange), Carneiro4_098 (Wage labor), Carneiro4_099 (Standardized medium of exchange), Carneiro4_100 (Coined money), Carneiro4_103 (Itinerant merchants or traders), Carneiro4_104 (Sedentary merchants), Carneiro4_105 (Markets), Carneiro4_106 (Daily and permanent markets), Carneiro4_112 (Trade between communities), Carneiro4_113 (Trade with other societies) (Carneiro, 2025). All SCCS and Carneiro data were retrieved from D-PLACE (Kirby et al., 2016).

TABLE A.4. CAPITALISM INDEX VALIDATION RESULTS

	Commercial economy (eHRAF)	Log exposure to ancient trade	Log distance to ancient trade	Log nighttime light (Old World)
Ethnographic capitalism index	1.693***	0.609***	-0.431***	0.179***
	(0.321)	(0.191)	(0.139)	(0.056)
Observations	346	324	324	184

Notes: * p < 0.1, ** p < 0.05, *** p < 0.01. Standard errors in parentheses. Each external measure of capitalism is regressed on the standardized residuals of the text-based capitalism index (obtained by regressing the index on the baseline controls used in the main analysis). The commercial economy measure is an indicator coded by eHRAF anthropologists. The measure of exposure and distance to ancient trade are derived from the global dataset of ancient trade routes and hubs described in in Appendix B. Average nighttime light within 100km of each culture is obtained from the Earth Observation Group's Annual VNL V2 series (Elvidge et al., 2021). Observations in the Americas and Oceania are excluded from the nighttime light regressions.

TABLE A.5. PROSOCIALITY INDEX VALIDATION RESULTS

	Ethnographic prosociality index	Generosity (SCCS)	Prosociality in experiments
		Panel A. Folklore	
Folklore prosociality index	0.092	-0.004	0.225
	(0.067)	(0.029)	(0.366)
Observations	288	103	20
		Panel B. Ethnography	
Ethnographic prosociality index		-0.004	-0.177
		(0.012)	(0.125)
Observations		102	15

Notes: * p < 0.1, ** p < 0.05, *** p < 0.01. Standard errors in parentheses. Each external measure of prosociality is regressed on the standardized residuals of the text-based prosociality index (obtained by regressing the index on the baseline controls used in the main analysis). Panel A reports the coefficient estimates for the folklore index, and Panel B reports the coefficient estimates for the ethnography index. The generosity measure is a derived from the variable SCCS334 (Generosity) (Barry et al., 1976), which was retrieved from D-PLACE (Kirby et al., 2016). The measure of prosociality in economic experiments is derived from a series of cross-cultural economic experiments (Henrich et al., 2004, 2010).

TABLE A.6. MORALITY INDEX VALIDATION RESULTS

	Ethnographic morality index	Moralizing high god (EA)	Christianization / Islamization (SCCS)
		Panel A. Folklore	
Folklore morality index	0.104*	0.120***	0.122***
	(0.059)	(0.020)	(0.032)
Observations	288	452	182
		Panel B. Ethnography	
Ethnographic morality index		0.064**	0.046
		(0.031)	(0.032)
Observations		245	184

Notes: * p < 0.1, ** p < 0.05, *** p < 0.01. Standard errors in parentheses. Each external measure of morality is regressed on the standardized residuals of the text-based morality index (obtained by regressing the index on the baseline controls used in the main analysis). Panel A reports the coefficient estimates for the folklore index, and Panel B reports the coefficient estimates for the ethnography index. The moralizing high god variable (derived from EA034 in the Ethnographic Atlas) indicates whether a high god is present, active in human affairs, and supportive of human morality (Murdock et al., 2025). The Christianization / Islamization variable (derived from SCCS2002) indicates whether cultures are deeply Christianized or Islamized (Rosenblatt et al., 2011). EA and SCCS data were retrieved from D-PLACE (Kirby et al., 2016).

TABLE A.7. ECOCENTRISM INDEX VALIDATION RESULTS

	Ethnographic ecocentrism index	Log human modification (Old World)
	Panel A. Folklore	
Folklore ecocentrism index	0.062	-0.004
	(0.061)	(0.005)
Observations	288	480
	Panel B. Ethnography	
Ethnographic ecocentrism index		-0.015*
		(0.007)
Observations		171

Notes: * p < 0.1, ** p < 0.05, *** p < 0.01. Standard errors in parentheses. Each external measure of ecocentrism is regressed on the standardized residuals of the text-based ecocentrism index (obtained by regressing the index on the baseline controls used in the main analysis). Panel A reports the coefficient estimates for the folklore index, and Panel B reports the coefficient estimates for the ethnography index. The human modification variable measures the degree to which humans have modified the ecosystems within 100km of each culture as of 2020 (Theobald et al., 2025). Observations in the Americas and Oceania are excluded from the human modification regressions.

TABLE A.8. STRATIFICATION INDEX VALIDATION RESULTS

	Ethnographic stratification index	Male bias in folklore	EA stratification	SCCS stratification	Carneiro stratification
			Panel A. Folklore		
Folklore stratification index	0.089 (0.070)	0.015*** (0.004)	0.029** (0.012)	0.060 (0.044)	0.022 (0.045)
Observations	288	943	532	182	113
			Panel B. Ethnography		
Ethnographic stratification index		0.012** (0.006)	0.122*** (0.016)	0.147*** (0.035)	0.178*** (0.048)
Observations		288	266	184	71

Notes: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses. Each external measure of stratification is regressed on the standardized residuals of the text-based stratification index (obtained by regressing the index on the baseline controls used in the main analysis). Panel A reports the coefficient estimates for the folklore index, and Panel B reports the coefficient estimates for the ethnography index. The male bias variable created by Michalopoulos and Xue (2021) measures the prevalence of misogynistic stereotypes in each folklore tradition. The EA stratification index is derived from EA066 (Class differentiation), EA068 (Caste differentiation), and EA 070 (Slavery) (Murdock et al., 2025). The SCCS stratification indicator is derived from the variable SCCS158 (Social stratification) (Murdock & Provost, 1973). The Carneiro stratification index is derived from the variables Carneiro4_088 (Wealth differences among society members), Carneiro4_121 (Extrasocietal slavery), Carneiro4_122 (Intrasocietal slavery), Carneiro4_123 (Slaves bought and sold within the society), Carneiro4_126 (Significant social status differences), and Carneiro4_127 (Social stratification into at least three classes) (Carneiro, 2025). All EA, SCCS, and Carneiro data were retrieved from D-PLACE (Kirby et al., 2016).

TABLE A.9. SUBSISTENCE TYPE INDEX VALIDATION RESULTS

	Ethnographic index	eHRAF indicator	EA index	SCCS index	Carneiro index
			Panel A. Agriculture		
Folklore agriculture index	0.188***	0.196	0.728***	0.066***	0.166***
	(0.052)	(0.123)	(0.115)	(0.022)	(0.052)
Observations	288	288	608	182	113
Ethnographic agriculture index		0.677***	1.612***	0.138***	0.230***
		(0.102)	(0.160)	(0.016)	(0.033)
Observations		346	286	184	71
			Panel B. Pastoralism		
Folklore pastoralism index	0.338***	0.774***	0.869***	0.091***	0.239***
	(0.085)	(0.241)	(0.087)	(0.017)	(0.049)
Observations	288	288	608	182	113
Ethnographic pastoralism index		2.720***	1.135***	0.128***	0.144***
		(0.512)	(0.242)	(0.015)	(0.035)
Observations		346	286	184	71
			Panel C. Hunting and gathering		
Folklore hunt/gather index	0.253***	0.607***	0.586***	0.069**	
	(0.068)	(0.144)	(0.089)	(0.028)	
Observations	288	288	608	182	
Ethnographic hunt/gather index		1.217***	1.248***	0.190***	
		(0.125)	(0.110)	(0.018)	
Observations		346	286	184	

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses. Each external subsistence type measure is regressed on the standardized residuals of the text-based subsistence type index (obtained by regressing the index on the baseline controls used in the main analysis). Panel A reports the coefficient estimates for agriculture indices, Panel B for pastoralism indices, and Panel C for hunting and gathering indices. The EA indices are derived from EA001-EA005 (Subsistence economy) (Murdock et al., 2025). The SCCS indices are derived from SCCS814-SCCS818 (Percent importance of agriculture / domestic animals / fishing / hunting / gathering in contribution to subsistence and trade) (Barry & Schlegel, 1982). The Carneiro index for agriculture is derived from Carneiro4_001 (Agriculture present) and Carneiro4_002 (Agriculture is primary subsistence), and the Carneiro index for pastoralism is derived from Carneiro4_024 (Domesticated animals kept to provide food) (Carneiro, 2025). All EA, SCCS, and Carneiro data were retrieved from D-PLACE (Kirby et al., 2016).

A.3. Robustness checks using external data

I employ the external cross-cultural data that I used to test the indices for validity in several robustness checks of the main results. Table A.10 reports the coefficient estimates from regressing external measures of prosocial enculturation, belief in moralizing high gods, anthropocentrism, and stratification on external measures of market exposure. In the SCCS sample, Christianization and Islamization and stratification increase with market exposure, but markets are unrelated to the cultural emphasis on teaching children generosity (Barry et al., 1976; Barry & Schlegel, 1982; Murdock & Morrow, 1970; Murdock & Provost, 1973; Rosenblatt et al., 2011). In the Carneiro

sample, markets are strongly correlated with stratification (Carneiro, 2025). Log exposure to ancient trade (Appendix B) is positively correlated with log human modification of ecosystems (Theobald et al., 2025) in the Old World. These results provide evidence that the relationships between market exposure and cultural salience of morality, anthropocentric ontologies and behavior, and stratified societies exist outside of folklore and ethnographic text.

Table A.11 reports the main OLS regression results for the subset of the folklore and ethnographic observations present in the Standard Cross-Cultural Sample with and without controlling for religion. The religion control is a categorical variable based on the SCCS variable "World Religions (1807)" (SCCS2002), which indicates the predominant religion in each culture (Indigenous religion, deep Islamization, deep Christianization, superficial Islamization, superficial Christianization, Mahayana Buddhism, Hinayana Buddhism, Vajrayana Buddhism, and Hinduism) (Rosenblatt et al., 2011). The social and cultural correlates of markets persist when controlling for religion, suggesting that the effects of trade networks on the spread of world religions is not driving the main results. I do not report the IV results because exposure to ancient trade is a weak instrument for the ethnographic markets index when restricting the estimation to the SCCS subset of the data.

TABLE A.10. MAIN REGRESSION RESULTS FROM EXTERNAL CROSS-CULTURAL DATA

	Generosity (SCCS)	**Christianization / Islamization (SCCS)**	**Log human modification (Old World)**	**Stratification (SCCS)**	**Stratification (Carneiro)**
Markets (SCCS)	0.023 (0.092)	0.507*** (0.109)		0.911*** (0.145)	
Log exposure to ancient trade			0.020*** (0.002)		
Markets (Carneiro)					1.127*** (0.085)
Observations	107	189	650	189	137

Notes: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses. These regressions mimic the main analysis but employ the external validity data. The SCCS market exposure index is a derived from the variables SCCS1 (Intercommunity trade as food source), SCCS155 (Money), and SCCS819 (Percent importance of trade in contribution to subsistence and trade) (Barry & Schlegel, 1982; Murdock & Morrow, 1970; Murdock & Provost, 1973). The Carneiro market exposure index is derived from the variables Carneiro4_020 (Regular food surplus production), Carneiro4_091 (Craft production for economic exchange), Carneiro4_098 (Wage labor), Carneiro4_099 (Standardized medium of exchange), Carneiro4_100 (Coined money), Carneiro4_103 (Itinerant merchants or traders), Carneiro4_104 (Sedentary merchants), Carneiro4_105 (Markets), Carneiro4_106 (Daily and permanent markets), Carneiro4_112 (Trade between communities), Carneiro4_113 (Trade with other societies) (Carneiro, 2025). The measure of exposure and distance to ancient trade are derived from the global dataset of ancient trade routes and hubs described in in Appendix B. The generosity measure is a derived from the variable SCCS334 (Generosity) (Barry et al., 1976). The Christianization / Islamization variable (derived from SCCS2002) indicates whether cultures are deeply Christianized or Islamized (Rosenblatt et al., 2011). The human modification variable measures the degree to which humans have modified the ecosystems within 100km of each culture as of 2020 (Theobald et al., 2025). The SCCS stratification indicator is derived from the variable SCCS158 (Social stratification) (Murdock & Provost, 1973). The Carneiro stratification index is derived from the variables Carneiro4_088 (Wealth differences among society members), Carneiro4_121 (Extrasocietal slavery), Carneiro4_122 (Intrasocietal slavery), Carneiro4_123 (Slaves bought and sold within the society), Carneiro4_126 (Significant social status differences), and Carneiro4_127 (Social stratification into at least three classes) (Carneiro, 2025). All SCCS and Carneiro data were retrieved from D-PLACE (Kirby et al., 2016).

TABLE A.11. ROBUSTNESS TO RELIGION CONTROL (SCCS)

	Prosociality	Morality	Ecocentrism	Stratification
		Panel A. Folklore indices (OLS)		
Markets	0.796***	1.179***	-0.426***	0.255**
	(0.124)	(0.150)	(0.110)	(0.121)
Observations	187	187	187	187
Religion control	No	No	No	No
Markets	0.482**	1.090***	-0.496***	0.149
	(0.188)	(0.257)	(0.157)	(0.094)
Observations	187	187	187	187
Religion control	Yes	Yes	Yes	Yes
		Panel A. Ethnographic indices (OLS)		
Markets	0.148***	0.015	-0.073***	0.278***
	(0.050)	(0.015)	(0.021)	(0.077)
Observations	187	187	187	187
Religion control	No	No	No	No
Markets	0.161***	0.003	-0.069***	0.216***
	(0.046)	(0.018)	(0.023)	(0.079)
Observations	187	187	187	187
Religion control	Yes	Yes	Yes	Yes

Notes: * p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors (in parentheses) are clustered by language family. Baseline controls include the number of motifs and its square, the number of sources, and the mean publication year of the source. For the IV regressions in Panel B, exposure to ancient trade is used as an instrument for the text-based market index. The religion control (SCCS2002) is a categorical variable based on the prevailing religion (Indigenous religion, Islam, Christianity, Buddhism, or Hinduism) (Rosenblatt et al., 2011); I retrieved it from D--PLACE (Kirby et al., 2016).

Appendix B. Historical market exposure

In this appendix, I describe the assembly of the ancient trade dataset used to construct the historical trade exposure instrument (Figure B.1), detail the derivation of the measure, and show that the IV results are robust to varying the range of trade influence (σ).

In "Trade and Geography in the Spread of Islam," Michalopoulos, Naghavi, and Prarolo (MNP) develop a dataset of historical trade routes and hubs in the Old World (2018). This dataset was subsequently employed by Michalopoulos and Xue (2021) to show that the prevalence of trade-related folklore motifs is associated with distance from historic trade routes in the Old World. In "Market Exposure and Human Morality," Enke uses distance from these historic trade routes as an instrument for market exposure in his global IV analysis of the cultural effects of markets, but

he does not supplement MNP's Old World data with any data on historical trade in the "New World" (2023).

I extend MNP's dataset to the whole world. I compile a list of the coordinates of ancient and early medieval trade routes and hubs from a number of sources, complement the pre-existing data in areas of sparse coverage, and filter the resulting global compendium of trade sites to eliminate duplicate observations. Six sources provide the majority of the observations:

1. Michalopoulos, Naghavi, and Prarolo (2018): I include the pre-600 CE trade sites.
2. Arthur de Graauw's catalogue of ancient ports (2022) (8th edition).
3. Pleiades gazetteer of ancient places (Elliott et al., 2025): I include all ports as well as places with "market," "trade," "trading," "emporium," or "caravan," in the description.
4. The Mapping Past Societies project (McCormick, 2026). I include six layers digitized from the *Barrington Atlas of the Greek and Roman World* (Talbert, 2000), *The Atlas of Medieval Europe* (Ditchburn et al., 2007), and *Grosser Historischer Weltatlas* (Bengtson & Milojcic, 1995):
    i. Hanseatic Traders Sites, ca. 1100-1299 CE
    ii. Important Fairs, ca. 900 – 1500 CE
    iii. Market Sites, pre 550 BCE – 640 CE
    iv. Mints, 800 – 1000 CE
    v. Ports and Harbors (Barrington), pre 550 BCE – 640 CE
    vi. Route and Station Points, pre 550 BCE – 640 CE)
5. The Old World Trade Routes project (OWTRAD) (Ciolek, 2012): I include all trade hubs from ancient and early medieval times as well as trade structures categorized as markets, qaysariyyas, or ports.
6. UNESCO World Heritage Sites: I include all ancient and early medieval sites related to markets and trade.

I add a number of trade sites across Africa, Asia, Oceania, and the Americas based on the *Atlas of World History* (O'Brien, 1999) and *The Complete Atlas of World History*, Volume 1 (Haywood, 1997). Lastly, I searched ancientlocations.net, a catalogue of archaeological sites, for trade- and market-related sites in the Americas. Many of the sites appear in several sources (e.g. Rome, Carthage, Alexandria, and major cities along the Silk Road), so I cluster observations that occur

within 20 km of one another and drop all but one in each cluster. The resulting dataset has 7,442 observations. Figure B.1 presents a map of these ancient trade routes and hubs.

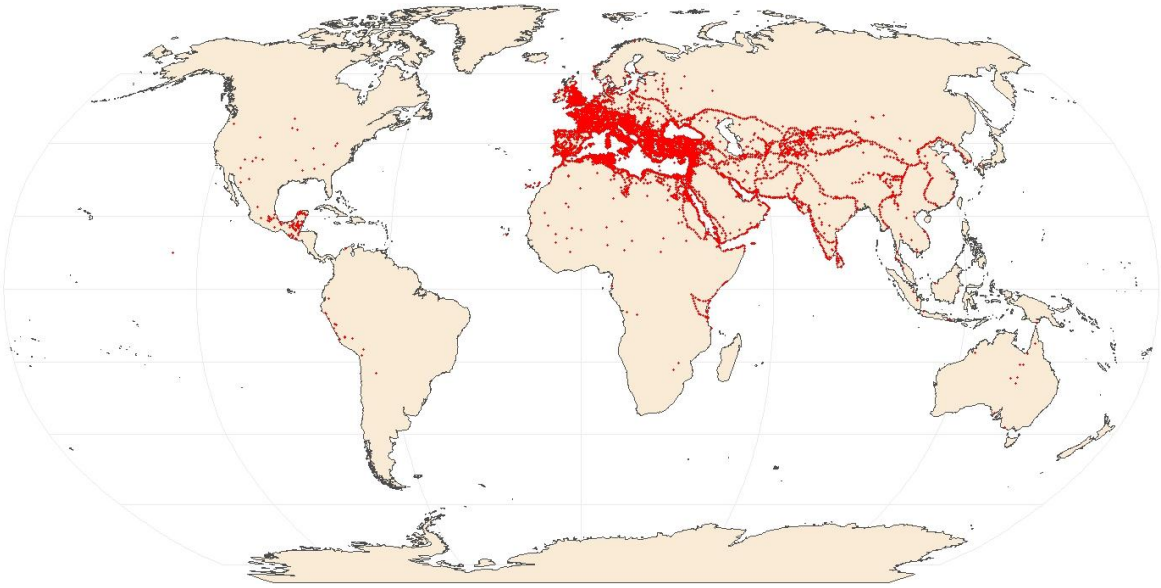


FIGURE B.1. MAP OF ANCIENT TRADE ROUTES AND HUBS

Notes: The ancient and early medieval trade routes and hubs marked on this map were used to construct the measure of historical trade exposure used as an instrument in the IV analysis. This dataset of trade sites was compiled from a number of sources, including Michalopoulos, Naghavi, and Prarolo's map of Old World trade (2018), the catalogue of ancient ports (de Graauw, 2022), the Pleiades gazetteer of ancient places (Elliott et al., 2025), the Mapping Past Societies project at Harvard (McCormick, 2026), the Old World Trade Routes project (OWTRAD) (Ciolek, 2012), the UNESCO World Heritage List, the *Atlas of World History* (O'Brien, 1999), and *The Complete Atlas of World History*, Volume 1 (Haywood, 1997).

Rather than using distance to the nearest trade site as the instrument (as Enke does), I develop a measure that captures the intensity of market exposure based on proximity to every trade site. I use a Gaussian function to transform distance between society and trade site into a number between 0 and 1. Then, for each society, the measure of historical trade exposure is obtained by summing across these Gaussian kernels for all trade sites:

$$log\ trade\ exposure_i = \ln\left(1 + \sum_{j\ \epsilon\ \text{sites}} \exp\left(-\frac{\text{dist}(i,j)^2}{2\sigma^2}\right)\right) \qquad (B.1)$$

The Gaussian function goes cleanly to 0, so trade sites that are very distant from a society do not influence its trade exposure values. Additionally, the Gaussian function does not go to infinity as the distance approaches 0 (as an inverse distance measure would), so small differences in the distance of nearby trade sites do not translate into disproportionately divergent contributions to the trade exposure measure. The Gaussian function I use to calculate historical trade exposure measure contains a parameter, σ, which represents the range of influence of trade. For my main analysis, I use σ = 1,000 km. Under this specification, trade sites more than 3,000 km away from a society

contribute negligibly to its trade exposure. To certify that this parameter choice is not driving the results, I estimate the IV regression coefficients of the social and cultural effects of markets for ten different values of σ, from 200 km to 2,000 km (Figure B.2). The results show that the coefficient estimates are reasonably stable in size and statistical significance across these different specifications of the historical trade exposure instrument.

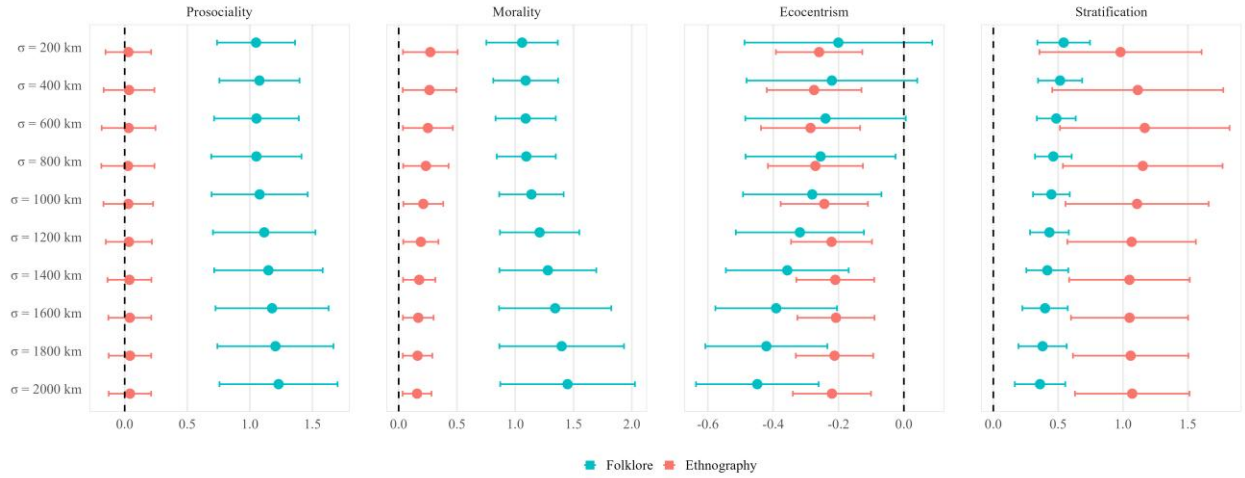


FIGURE B.2. IV ESTIMATES FROM DIFFERENT RANGES OF TRADE EXPOSURE

Notes: This figure reports the IV coefficient estimates from regressing prosociality, morality, ecocentrism, and stratification on markets, varying the range of trade exposure (σ) from 200 km to 2,000 km (σ = 1,000 km in the main analysis). Coefficient estimate error bars represent 95% confidence intervals. All regressions include baseline controls.